SMART MOVIE RECOMMENDATION SYSTEM USING MACHINE LEARNING

1. Introduction

Recommendation systems are widely used in modern digital platforms such as Netflix, Amazon Prime Video, and Spotify to personalize user experience. These systems analyze user preferences and content similarities to suggest relevant items.

This project implements a Content-Based Movie Recommendation System using Natural Language Processing (NLP) techniques. The system recommends movies based on textual similarity between genre and description features.

2. Abstract

The objective of this project is to build an AI-powered movie recommendation system using TF-IDF vectorization and cosine similarity. The model processes movie metadata such as genre and description, converts textual data into numerical vectors, and calculates similarity scores between movies.

Based on these similarity scores, the system recommends the top N most similar movies selected by the user. The application is deployed using Streamlit with a modern and interactive user interface.

3. Tools & Technologies Used

- Python
- Pandas
- Scikit-Learn
- Streamlit
- TF-IDF Vectorizer
- Cosine Similarity
- VS Code

4. Dataset Description

A custom dataset containing more than 60 movies was created. Each movie includes:

- Title
- Genre (single or multi-genre)
- Short description

The dataset includes diverse categories such as:

- Action
- Sci-Fi
- Drama
- Thriller
- Fantasy
- Horror
- Animation
- Biography
- War
- Crime

This diversity improves recommendation quality and system robustness.

5. Methodology

Step 1: Data Preprocessing

- Combined genre and description into a single content column.
- Removed stopwords using TF-IDF configuration.
- Structured textual data for vectorization.

Step 2: Feature Engineering

TF-IDF (Term Frequency – Inverse Document Frequency) was applied to convert text into numerical vectors. This technique assigns higher weight to important words while reducing the impact of common words.

Step 3: Similarity Calculation

Cosine similarity was used to measure similarity between movie vectors. The similarity score ranges from 0 to 1:

- 1 → Highly similar
- 0 → Not similar

Step 4: Recommendation Logic

- Identify selected movie index
- Compute similarity scores
- Sort in descending order
- Return Top-N similar movies

Step 5: Deployment

The system was deployed using Streamlit with:

- Sidebar controls
- Dynamic movie selection
- Adjustable number of recommendations

- Styled UI components
- Similarity score visualization

6. Results

The system successfully recommends movies based on content similarity. For example:

- Input: “Inception”
- Output: “Interstellar”, “The Matrix”, “Dune”, etc.

The recommendation quality improves due to multi-genre combination and descriptive features.

7. Conclusion

This project demonstrates practical implementation of machine learning and NLP techniques in building a real-world recommendation system. It highlights how TF-IDF and cosine similarity can be applied effectively for content-based filtering.

Future improvements can include:

- Hybrid recommendation model
- Collaborative filtering
- Integration with real-world APIs
- Cloud deployment

The project successfully meets its objective and provides a strong foundation for advanced recommendation system development.